

A Polarization Reconfigurable Circular Patch Antenna with Tuning Stub and Crescent Slot

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Abstract—A polarization-reconfigurable patch antenna designed for future 5G application operating in the ka band is introduced. This study focuses on designing and analyzing a circular-shaped microstrip patch antenna, utilizing a Rogers RT Duroid 5880 substrate, to achieve diversity polarization. The antenna is designed with dimensions of 24 mm × 20 mm × 0.80 mm. The antenna consists of a circular radiating element with crescent-shaped slots, fed by a microstrip transmission line, and incorporates two stubs, each integrated with independently controlled PIN diodes. The polarization can be toggled between right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP) by switching the states of the PIN diodes. Simulation results show that the antenna operates between 26.63 GHz and 29.76 GHz with a reflection coefficient below -10 dB. The antenna exhibits a gain exceeding 6 dBi across all polarization states throughout the entire designed frequency range. Additionally, it achieves a good axial ratio (less than 3 dB) in both RHCP and LHCP modes over the 27.82-28.36 GHz range and acquires axial-ratio bandwidth (ARBW) of 1.82%. Its simple design and ease of simulated make it an excellent candidate for high-speed data transmission and 5G ka band applications.

Index Terms—Ka Band, 5G, LHCP, RHCP, Axial ratio, ARBW, CST.

I. INTRODUCTION

In the contemporary age, the rapid growth of 5G wireless communication has emerged as a transformative technology, promising to revolutionize various industries and daily life. To meet the increasing demand for higher data rates, lower latency, and massive connectivity, 5G systems utilize in higher frequency bands, including the Ka band (26.5-40 GHz) [1], [2]. Microstrip patch antennas are extensively used in 5G communications because of their advantages, including a low profile, lightweight design, and ease of fabrication. However, conventional single-polarized antennas may fall short in addressing the increasing demand for multiple communication standards and polarization diversity [3]. Dual-polarized [4] antennas, capable of supporting both vertical and horizontal polarizations, offer increased channel capacity, improved signal reception, and greater resistance to multipath fading, making them ideal for 5G applications.

To further increase the flexibility of the antenna, Reconfigurable antennas are highly favored by antenna designers than normal microstrip patch antenna and manufacturers because they offer multi-functional capabilities within a single antenna design. These antennas can adapt to changes in frequency, radiation pattern, polarization, and operating bands through

various innovative techniques [4]–[8]. Additionally, Antennas capable of polarization switching are highly desirable for both wireless and satellite communication systems. Polarization reconfiguration also provides substantial flexibility by enabling the use of either polarization uniformity or polarization diversity. This includes the ability to switch between linear polarization (LP) in both vertical and horizontal orientations, as well as between right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP). This adaptability enhances performance in various communication systems by optimizing signal reception and reducing interference.

Numerous antenna designs have been proposed that enable switching from linear to circular polarization (CP). To enable reconfiguration, switches like PIN diodes or microelectromechanical systems (MEMS) are commonly used to handle the operational state of the radiators, facilitating adjustments in polarization or beam direction as needed for dynamic applications [8]–[11].

Initially in 2016, S. W. Cheung and team [11] explore PIN diodes controlled to enable circular polarization switching. But, the antenna has an increased complexity in achieving efficient polarization switching. It does not operate higher frequency band. In the same year, M.A. Rahman and his mate proposed [12] a microstrip antenna with circular polarization switching for X-band Applications. The antenna design employs a truncated corner square patch with diodes to enable circular polarization switching between LHCP and RHCP by controlling the current applied to the diodes. However, this antenna may face challenges at higher 5G frequency bands like the Ka band and mm wave frequencies range. In recent years, M.D. Rheza et al. [13] proposed that the design employs a truncated-slot patch technique combined with control switches using PIN diodes to enable polarization diversity at the operating frequency of 3.5 GHz.

Although promising, at lower frequencies, these designs may still struggle at higher frequencies such as the Ka band and the mmWave range, where smaller wavelengths exacerbate challenges in material losses, manufacturing precision, and maintaining efficiency. The present used frequency band utilized for cellular networks is congested, leaving limited resources to support the high capacity needs of 5G technology [5], [14].

This study introduces a circular reconfigurable polarization antenna with tuning stubs and crescent slot microstrip patch

antenna for 5G ka band and mm-wave applications. Initially, the proposed antenna is simulated and the results are analyzed to determine whether the antenna is properly matched. Then, circular polarization can be achieved by incorporating a geometric feature, such as a stub, at the edge of the patch structure and incorporating a crescent slot. Two stubs are integrated into the geometric structure at a right angle to each other, with a PIN diode added to the stubs for biasing voltage switching functionality. The incorporation of stub and PIN diode segments introduces perturbations that facilitate the generation of circular polarization (CP). By adjusting the biasing voltage and toggling the diodes between the ON and OFF states, the antenna can dynamically switch between left-hand circular polarization (LHCP) and right-hand circular polarization (RHCP), enabling reconfigurable polarization characteristics. The goal of this work lies in the novel design and analysis of the antenna, which leverages geometric modifications and PIN diodes to achieve dynamic polarization reconfiguration, making it a suitable candidate for 5G Ka-band and mm-wave systems requiring adaptive and efficient polarization control.

The following sections are organized as outlined below: Section II details the design process and operating principle of the proposed dual-polarization reconfigurable patch antenna. Section III presents the simulation results, experimental findings, and discussion. Finally, Section IV concludes the study.

II. ANTENNA DESIGN PROCEDURE

A. Proposed Antenna's Geometry

A circular shaped patch is placed on the top of the 0.80 mm thickness Rogers RT Duroid 5880 substrate that exhibits a relative dielectric constant (ϵ) of 2.20 above the copper-annealed ground layer. The proposed patch is excited by a microstrip feed line and the two stub is added at the edge of the the circular patch. D1 and D2 PIN diodes are set up in between the two stubs and the edge of the circular patch for polarization switching. The geometrical layout of dual polarization reconfigurable antenna is illustrated in Fig. 1. The dimension of the proposed single element ground is 24×20 mm² shown in Fig. 2. The overall dimension and parameter of the proposed antenna is presented in Table I.

B. Operating Principle of Proposed Antenna

This section contains the working principle of proposed reconfigurable circular polarization antenna. Initially, the antenna shows linear polarization operation shown in Fig. 1 (a). The crescent-shaped slot with an additional stub embedded in the circular patch modifies the current distribution and helps in enhancing the antenna's bandwidth and polarization properties. This slot perturbs the symmetry of the circular patch and enhance the circular polarization properties. This process help the antenna design to introduce two orthogonal modes with a right angle phase shift to achieve circular polarization.

The antenna's reconfigurability is achieved through two independently controlled PIN diodes. There is a biasing voltage that is used to control the PIN diode. The equivalent model of a PIN diode is given in Fig. 4. By selectively biasing these

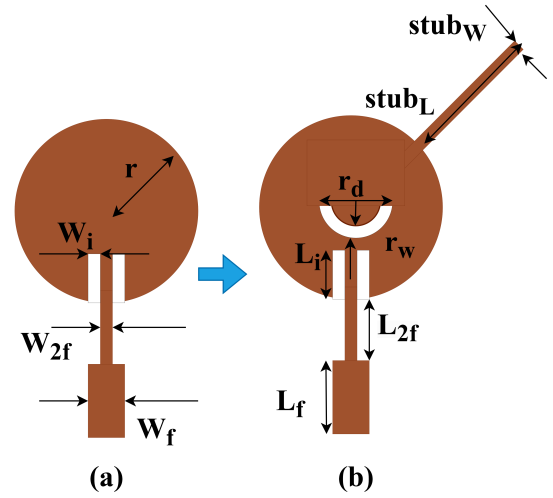


Fig. 1: Proposed circular patch: (a) Circular patch without substructure, (b) Circular patch with stub.

TABLE I: Designing parameters corresponding to the Proposed Reconfigurable antenna.

Design Parameters	Values (mm)
Substrate Length (L_s)	24
Substrate Width (W_s)	20
Patch radius (r)	3.34
Feed Line Width (W_f)	0.62
Feed Line Length (L_f)	1.82
Transformer Length (L_{2f})	2.98
Transformer Width (W_{2f})	2.49
Inset feed Length (L_i)	1.40
Inset feed Width (W_i)	0.08
Half ring slot width (r_w)	0.25
Slot diameter (r_d)	1.45
Stub width ($stub_W$)	0.25
Stub length ($stub_L$)	8.9 (from center of patch radius)
Stub angle ($angle$)	45°

diodes, the current distribution on the patch is altered, enabling the antenna to switch between right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP) which is illustrated in Fig. 2.

PIN diodes function as switches in the proposed antenna. The characteristics and performance of the PIN diode are affected by resistors (R) and the applied biasing voltage (V) which is depicted in Fig. 3. This configuration enables switching between the ON and OFF states by controlling the PIN diodes.

When a forward-bias voltage is applied to the PIN diode, it switches to the ON state, while a reverse-bias voltage places it in the OFF state. The circular patch is shorted with stub in ON state and isolated in another stub in OFF state shown in Fig. 5. By switching the both diodes ON or OFF, both senses of circular polarization can be accomplished, specifically right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP). By activating D1 diode or deactivating D2 diode elements, it allows current to flow through a specific

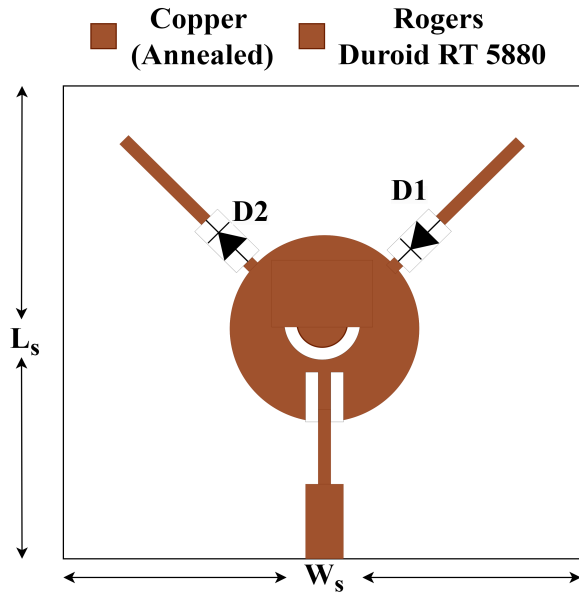


Fig. 2: Geometric layout of proposed antenna.

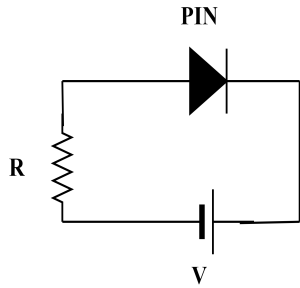


Fig. 3: PIN diode with biasing voltage.

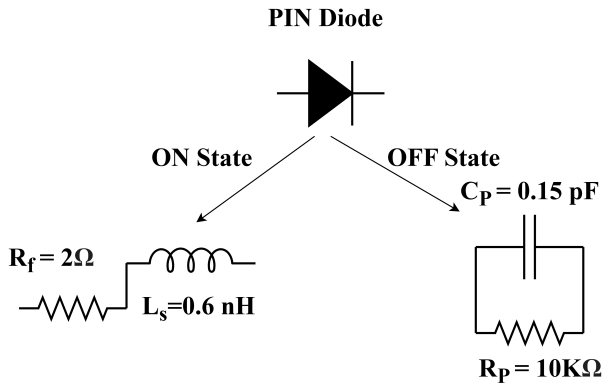


Fig. 4: Equivalent model of a PIN diode.

pattern, enabling left-hand circular polarization (LHCP) of the proposed antenna. Conversely, by activating D2 diode and deactivating D1 diode, the current distribution changes, allowing the antenna to switch to the opposite circular polarization like right-hand circular polarization (RHCP).

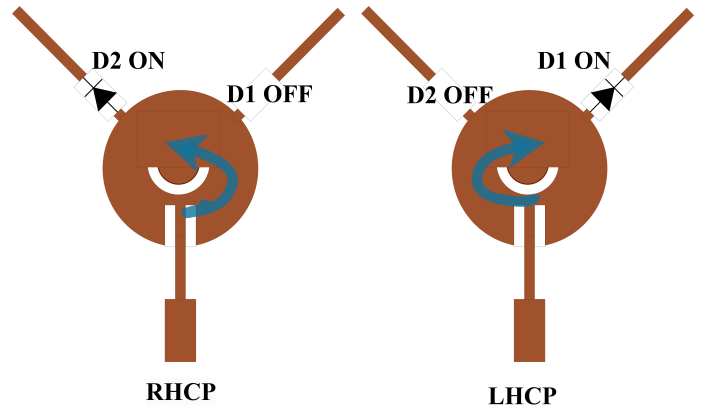


Fig. 5: Working principle of the proposed reconfigurable antenna for circular polarization operation.

III. SIMULATED RESULT AND DISCUSSION

A. Reflection Coefficient and Gain of proposed antenna

Fig. 6 illustrates the reflection Coefficient of the proposed antenna under various conditions. The graph clearly demonstrates that the antenna is well-matched within the frequency range of 26.63 GHz to 29.75 GHz. The antenna achieves a minimum S_{11} value of approximately -31.41 dB at a frequency of 28.02 GHz. The antenna achieves a peak impedance bandwidth of 11.11%, measured at the -10 dB threshold.

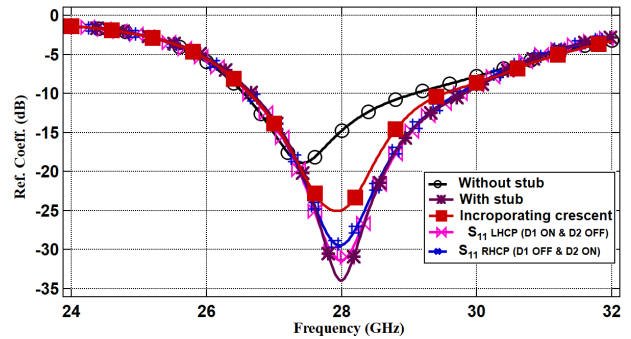


Fig. 6: Simulated reflection coefficient of proposed antenna.

The gain pattern of the proposed microstrip antenna, as depicted in Fig. 7, demonstrates a consistently strong performance, with a gain exceeding 6 dBi across the entire frequency range.

Table II provides an overview of the antenna's reconfigurability in terms of frequency range, bandwidth polarization state, axial ratio and axial ratio bandwidth based on the different switching states of the PIN diodes by applying biasing voltage.

B. Axial Ratio of proposed patch antenna

The axial ratio curve for the proposed antenna with various diodes is illustrated in Fig. 8. The figure shows the axial ratio under the different conditions that could lead to circular polarization (CP). As evident from the graph, the axial ratio in

TABLE II: The switching configuration of the proposed antenna controls the operating frequency and bandwidth adjustments.

Bias Voltage	D1 diode	D2 diode	State	Frequency range (GHz)	Resonant Frequency (GHz)	Bandwidth (%)	Axial Ratio (dB)	ARBW (%)
Forward	ON	OFF	LHCP	26.63 to 29.76	28.08	10.44	1.01	1.79
Reverse	OFF	ON	RHCP	26.63 to 29.72	28.10	10.98	0.80	1.82

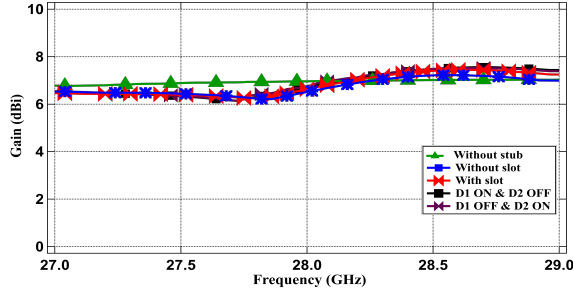


Fig. 7: Simulated gain versus frequency curve of the proposed patch antenna for several diode and analyzed polarization.

all the cases remains below 3 dB, indicating effective circular polarization and the ability to switch between polarization states like LHCP and RHCP.

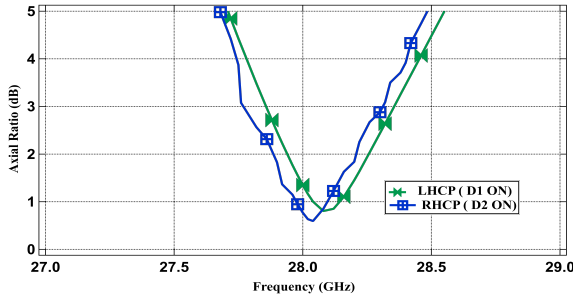


Fig. 8: Simulated axial ratio for several diode reconfiguration.

When both diodes are OFF state condition, then it shows linear polarization entire the bandwidth.

C. Parametric analysis of proposed antenna

A parametric study is being conducted on the proposed antenna to assess the impact of circular polarization. The analysis should be structured around the length, width and angle of the stub. It explains how each of these parameters affects the axial ratio (AR).

In Fig. 9, The axial ratio for left-hand circular polarization (LHCP) was simulated using the antenna's configuration, which includes the stub and crescent-shaped slots in the radiating element. This simulation was conducted without incorporating the PIN diodes, meaning the diodes were not activated or included in the design during this specific analysis. The objective was to evaluate the axial ratio performance of the antenna's inherent structure, focusing solely on the effects of the stub and crescent slots on circular polarization without the influence of the switchable elements.

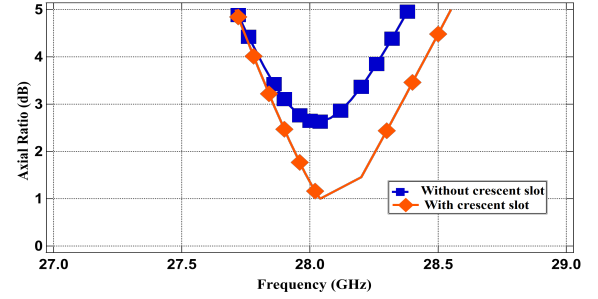


Fig. 9: The simulated axial ratio for LHCP was obtained with the stub and crescent slot configuration, excluding the use of PIN diodes.

Fig. 10 demonstrates that the length of the stub affects the circular polarization. With a stub length of 8.9 mm, the antenna can propagate circular polarization at the frequency of 28.1 GHz, which is a frequency band utilized in mm-Wave applications.

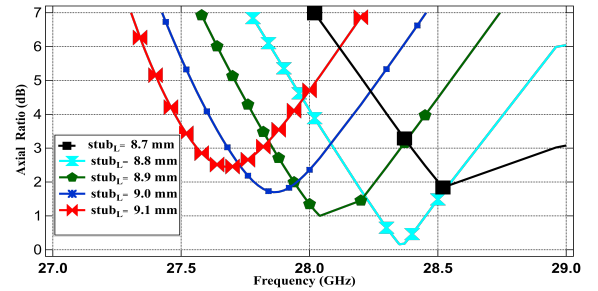


Fig. 10: Simulated axial ratio for LHCP for several stub length when D1 is turned ON and D2 is turned OFF.

The width of the stub also plays a significant to achieve circular polarization. By adjusting the stub width, it can be enhanced the performance of CP as well as axial bandwidth shown in Fig. 11.

The stub angle, a crucial parameter affecting the circular polarization of the proposed antenna, undergoes significant changes after optimization, as shown in Fig. 12.

D. Surface current

The polarization of the antenna alteration is achieved through the effective switching of two PIN diodes, which are integrated into the stub of the radiating structure. To observe the polarization behavior, the surface current vector distribution is analyzed at various phase angles ($\phi = 0^\circ, 90^\circ, 180^\circ$, and 270°). The consistent clockwise rotation of the surface current vectors at $\phi = 0^\circ$ through $\phi = 270^\circ$ indicates the

TABLE III: A comparison between the proposed antenna and several previously reported works.

References	No of switching elements	AR Bandwidth Range (GHz)	Axial Ratio Bandwidth	Axial Ratio (dB)	Antenna Gain (dB)
[5]	4	2.37-2.47 (LHCP) 2.36-2.46 (RHCP)	100 MHz	0.50	2.25 to 2.99 (LHCP) 2.1 to 2.98 (RHCP)
[6]	2	5.85-5.95	110 MHz	≤ 1	5 to 7.4
[11]	4	5.5-5.91	410 MHz	≤ 3	2.6 to 4.3
[13]	4	3.45-3.55	108 MHz	1.3 (LHCP) 1.283 (RHCP)	6.95
[14]	2	27.65-28.35	700 MHz	≤ 1.5	-
Proposed	2	27.86-28.36 (LHCP) 27.82-28.33 (RHCP)	500 MHz (LHCP) 505 MHz (RHCP)	0.80 (LHCP) 0.59 (RHCP)	6.45 to 7.30 (LHCP) 6.4 to 7.30 (RHCP)

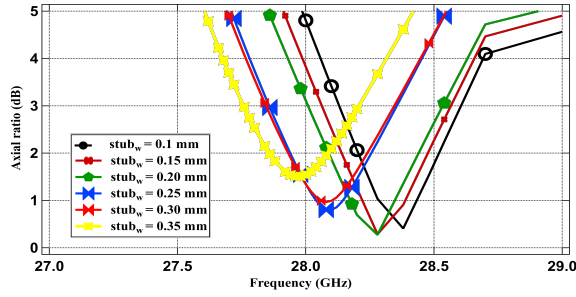


Fig. 11: Simulated axial ratio for LHCP for several stub width.

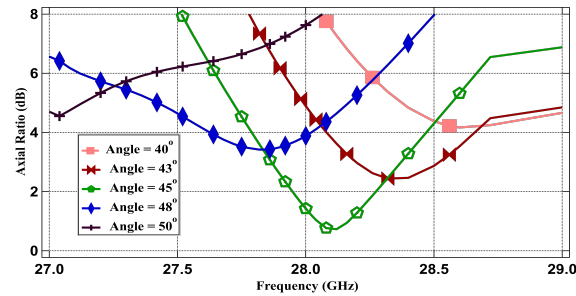


Fig. 12: Simulated axial ratio for LHCP for several stub angle.

generation of left-hand circular polarization (LHCP) radiation that is depicted in Fig. 13.

As observed in Fig. 14, the surface current vector distributions at phase angles ($\phi = 0^\circ, 90^\circ, 180^\circ$, and 270°) exhibit a consistent counterclockwise rotation. This progression indicates the generation of right-hand circular polarization (RHCP) radiation.

E. Comparison

In Table III provides an overview of the proposed reconfigurable antenna in comparison with previously published designs in the literature. The table presents a comprehensive summary of the key characteristics of the proposed reconfigurable antenna, including the number of switching elements, antenna gain axial ratio bandwidth (ARBW), and axial ratio performance. The comparison highlights the advantages of the proposed antenna in terms of Axial ratio, AR bandwidth, antenna gain etc.

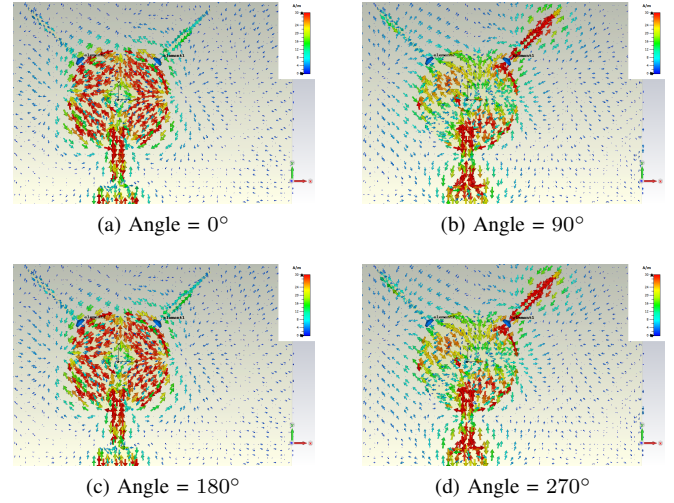


Fig. 13: Surface current distribution at varying phase angles demonstrates LHCP.

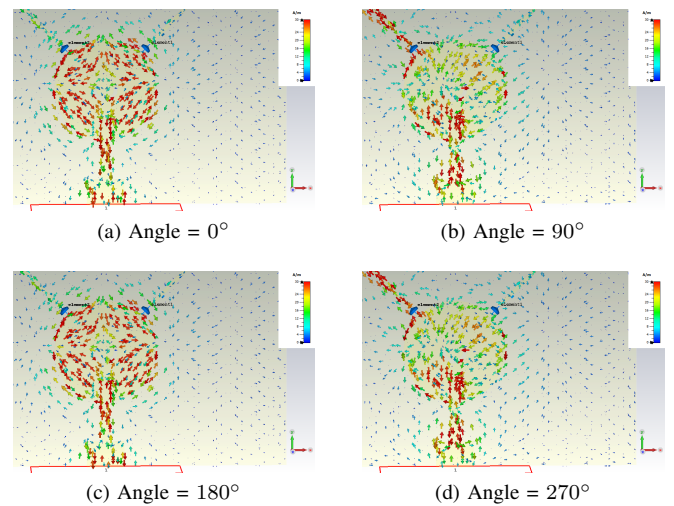


Fig. 14: Surface current distribution at varying phase angles demonstrates RHCP.

CONCLUSION

The paper examines the performance of a reconfigurable polarization microstrip patch antenna, which operates at 28.1 GHz for applications of the 5G Ka band, by adjusting the length, width, and angle of the stub. The antenna comprises a circular radiating element fed through a microstrip transmission line and includes two stubs with a crescent slot. Each equipped with independently controlled PIN diodes, allowing for left-hand or right-hand circular polarization. The simulation results indicate that the antenna demonstrates perfectly matching throughout the 26.63–29.76 GHz frequency range, with a 3 dB axial ratio achieved between 27.82 and 28.36 GHz for both polarization states. These characteristics make the proposed antenna well-suited for 5G ka band applications. The proposed antenna's capability to dynamically adjust its polarization makes it an excellent fit for advanced 5G Ka-band communication systems. It is particularly beneficial for applications that demand improved signal quality, minimized interference, and optimized spectrum utilization. Potential use cases include satellite communications, millimeter-wave wireless networks, and beam-steering systems, where adaptable and reconfigurable polarization is essential for reliable performance in changing environments.

IV. ACKNOWLEDGEMENT

The authors would like to thank Development and Research Extension (DRE) under Chittagong University of Engineering and Technology (CUET) for providing support through CUET/DRE/2023-24/ETE/016 project.

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